

RAGHUL S

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EDUCATION

Chennai Institute of Technology

B.Tech in Information Technology; CGPA: 8.2/10

Chennai, India

2023 – Present

EXPERIENCE

Software Development Intern

August 2024 – March 2025

Gogosoon Pvt. Ltd.

- Engineered **data collectors** for U.S. and Indian indices, handling high-concurrency ingestion pipeline with 100% integrity.
- Spearheaded backtesting infrastructure strategy by translating complex research papers into executable trading models.
- Orchestrated the migration of **3.4M legacy records** to production with **zero downtime**, ensuring rollback capabilities.
- Directed technical design reviews and code audits to **enforce test coverage** and architectural consistency.

PROJECTS

Parametric Yantra Generator

October 2025 – December 2025

- Integrated **Astronomy Engine** to compute **topocentric coordinates**, powering precise **3D sky simulations** of celestial bodies.
- Extracted **CAD blueprints** from **Three.js parametric models**, calculating dimensions via **latitude/longitude**.
- Implemented **shadow simulation** by mapping astronomical data to **Three.js light vectors** to validate instrument accuracy.
- Formulated **quantity surveying algorithms** to compute material volumes and construction metrics from **complex geometry**.

Exam Proctor Platform

July 2025 – September 2025

- Architected video recording system handling **100+ concurrent streams**, cutting storage costs **60-75% via H.264** encoding.
- Overhauled streaming architecture from sockets to **WebRTC**, **reducing latency 40%** to enable real-time client interaction.
- Enhanced face verification pipeline to reduce **server load 35%** while maintaining **95% detection accuracy** via model tuning.
- Implemented **SEB configurations** to secure exams with granular access control, achieving **100% process integrity**.

iHirn Cognitive Platform - Gogosoon

November 2024 – March 2025

- Developed core features for cognitive platform serving **1000+ users**, resolving critical bugs to maintain **99% uptime**.
- Maximised **cross-platform responsiveness** across desktop, iOS, Android by debugging platform-specific issues and testing.
- Upgraded game interface based on user feedback, deploying **adjustable difficulty levels** via **agile development cycles**.

Quantitative Trading System - Gogosoon

August 2024 – November 2024

- Constructed a high-frequency data pipeline processing **190K+ daily records**, using automated reliability monitoring systems.
- Boosted system performance by **76% latency reduction** through async I/O, ensuring **scalability** and sub-second responses.
- Designed a fully asynchronous backtesting engine achieving **8x performance boost**, enabling faster algorithm iteration.
- Devised data visualization dashboards for market analysis and built a testing framework to evaluate model performance.

TECHNICAL SKILLS

- Programming Languages:** JavaScript, TypeScript, Python, Go, C/C++, SQL, Java, Bash, HTML, CSS, Lua
- Technologies & Tools:** Git, Docker, Linux, AWS, REST API, MySQL, PostgreSQL, MongoDB, Jira, WebSocket, PGVector
- Frameworks & Libraries:** Node.js, React.js, Next.js, Express.js, Pandas, WebRTC, Matplotlib, Seaborn, LangChain, Three.js
- Development Practices:** Code Review, Multi-Threading, Async Programming, API Design, System Monitoring, CI/CD
- CS Fundamentals:** Data Structures, Algorithms, Computer Networks, Operating Systems, DBMS, OOP

COMPETITIVE PROGRAMMING

- LeetCode (RaghulSub):** Achieved rating of **1914**, **Top 4.71%**, solved **1115 problems**, and secured highest **rank 553** in contests
- CodeChef (raghulsub)** – Attained max rating of **1654 (3-star)** and secured peak **global rank 120** in Weekly challenges
- CodeForces (RaghulSub)** – Reached max rating of **1397 (Pupil)** and achieved top **global rank 2966** in competitive rounds

ACHIEVEMENTS

- Winner (Smart India Hackathon 2025)** - Engineered a geospatial calibration tool for the **Indian Knowledge Systems (IKS)** division to mathematically model and **dynamically** generate **precise** dimensions for ancient astronomical instruments.
- ICPC Prelims** - Solved **3 problems** and secured **global rank 1344** in the ICPC Asia-Pacific preliminary round
- Finalist (COD-a-thon)** - Competitive programming challenge, **ranked among the top 20** participants out of 200+ entrants

CERTIFICATIONS

- CCNA (CISCO):** Covered networking fundamentals, switching, IP connectivity, security fundamentals, and network automation.
- AWS Cloud Practitioner (CLF-C02):** Validated core cloud concepts, security, services, and billing practices.
- Introduction to Cybersecurity (CISCO):** Covered threat landscapes, network security principles, and data protection.